

Project: Agentic for Autonomous Research Outreach and Cold Email Automation

1. Input Definition

1.1 Primary Input (High-Level Natural Language Goal)

The system accepts an unstructured natural language instruction such as:

“I want to cold mail professors for a research internship.”

This input is:

- Ambiguous
- Incomplete
- Missing constraints
- Lacking structured parameters

The system must transform this vague request into a structured and executable task specification.

1.2 Missing Parameters to be Identified

The agent must infer or clarify the following:

- Objective (research internship, collaboration, industry outreach, etc.)
- Research domain(s)
- Geographic preference
- Target institutions (optional)
- Timeline
- User background (CV, projects, publications)
- Email tone preference
- Follow-up policy
- Outreach scale (number per week, per month)

These will be converted into structured state variables.

2. Intent Understanding and Clarification Module

2.1 Intent Extraction

The system must:

- Classify the high-level goal
- Extract available entities from input
- Detect missing required slots

This can be implemented using:

- LLM-based intent classification
- Named entity extraction
- Structured state tracking

Example internal representation:

```
{  
  "goal": "research_internship",  
  "domain": null,  
  "countries": null,  
  "timeline": null,  
  "user_profile": null  
}
```

2.2 Interactive Clarification

The system must operate as a slot-filling conversational agent.

It should:

- Identify missing fields
- Ask targeted follow-up questions
- Update the internal structured state after each response
- Confirm final structured objective before execution

Example:

User: "I want to cold mail professors."

Agent:

- “Which research domains are you interested in?”
- “Any specific countries?”
- “What timeline are you targeting?”
- “Please provide your CV.”

The clarification stage ends only when all mandatory parameters are filled.

3. Planning Module

Once sufficient information is collected, the agent generates a structured execution plan.

The plan must include:

- Search keywords
- Country filters
- Institution filtering strategy
- Relevance criteria
- Email drafting strategy
- Sending schedule
- Logging format

Example plan structure:

```
{
  "search_strategy": {
    "keywords": ["machine learning", "computer vision"],
    "countries": ["USA", "Germany"]
  },
  "filtering_criteria": {
    "publication_recency": true,
    "active_lab": true
  },
  "email_strategy": {
    "personalization_level": "high",
    "reference_publications": true
  }
}
```

The plan must be presented to the user for confirmation before execution.

4. Web Crawling and Information Extraction Module

4.1 Web Crawling

The agent must crawl:

- University department pages
- Lab websites
- Scholar pages
- Public academic directories

The crawler must retrieve structured information relevant to faculty members.

4.2 Professor Identification

For each candidate, the system extracts:

- Name
- Email address
- Department
- Research keywords
- Recent publications
- Indications of accepting students

4.3 Relevance Scoring

Each professor is assigned a relevance score based on:

- Keyword similarity
- Embedding similarity between user interests and professor publications
- Geographic match
- Recency of research output

Top-K professors are shortlisted for outreach and the agent will evaluate based on the raw data and this input as well

5. Personalized Email Generation

For each shortlisted professor:

The agent must:

- Analyze the professor's webpage
- Extract current research themes
- Identify recent papers
- Compare with the user's profile
- Generate a personalized email

The email must include:

- Specific reference to the professor's work
- Clear alignment with user's interests
- Concise description of the applicant
- Professional tone
- Attached CV

6. Supervised Execution Phase

In the initial deployment:

- The system shows:
 - Shortlisted professors
 - Extracted research summaries
 - Generated emails
- The user must approve each email before sending.
- The user can:
 - Edit content
 - Modify tone
 - Remove recipients
 - Adjust plan

This phase ensures correctness and builds trust.

7. Autonomous Execution Phase

After user approval of workflow:

The agent may operate independently.

It must:

- Periodically crawl for new matches
- Avoid duplicate outreach
- Track sent emails
- Monitor replies
- Send automated follow-ups after predefined intervals

This requires:

- Email API integration
- Scheduling engine
- Persistent database for tracking
- Duplicate detection logic

8. Workflow

The end-to-end workflow is defined as follows:

- Step 1: User provides vague high-level goal.
- Step 2: Agent extracts intent and identifies missing parameters.
- Step 3: Agent asks structured follow-up questions.
- Step 4: Agent builds structured state representation.
- Step 5: Agent generates and displays execution plan.
- Step 6: User approves plan.
- Step 7: Agent crawls web sources.
- Step 8: Agent extracts candidate professor data.
- Step 9: Agent ranks candidates based on relevance.
- Step 10: Agent drafts personalized emails.
- Step 11: User supervises first execution.
- Step 12: Emails are sent.
- Step 13: Agent tracks replies and sends follow-ups.
- Step 14: After approval, system transitions to autonomous mode.

Note that this is the tentative workflow, we might change it during development